Introduction to Electrical Engineering
Practice Exam

Discussion: January 12, 2026

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1 Conceptual Understanding: Multiple Choice

Select exactly one answer per question.

Fundamentals

1. The drift motion of electrons in a conductor is mainly caused by:

A) temperature gradients

B) magnetic fields

C) the electric field

D) mechanical forces

Solution: C

2. For a metallic conductor, an increase in temperature usually leads to:

A) decreasing resistance

B) sudden insulating behavior

C) unchanged resistance

D) increasing resistance

Solution: D

DC

3. Kirchhoff's current law states:

A) The sum of voltages in a loop is zero.

B) The sum of currents at a node is zero.

C) The sum of resistances at a node is constant.

D) Power is stored at a node.

Solution: B

4. An ideal voltage source is characterized by:

A) a fixed voltage independent of current

B) a fixed current independent of the load

C) infinite internal resistance

D) no influence on connected circuits

Solution: A

5. The nodal-voltage method primarily works with:

A) mesh currents

B) node voltages

C) equivalent sources

D) frequency analysis

Solution: B

6. For maximum power transfer from a source to a purely resistive load:

A) load resistance $\gg$ source resistance

B) load resistance $\ll$ source resistance

C) load resistance equals source internal resistance

D) load resistance equals zero

Solution: C

7. An ammeter is typically connected:

A) in parallel with the measured quantity

B) in series with the measured quantity

C) with zero internal resistance

D) only in AC circuits

Solution: B

Fields & Components

8. A capacitor primarily stores:

A) magnetic energy

B) mechanical energy

C) thermal energy

D) electric energy

Solution: D

9. Under constant DC voltage, a capacitor behaves (in steady state) like:

A) a short circuit

B) an open circuit (no current flow)

C) a resistor with fixed value

D) a voltage source

Solution: B

10. Magnetic fields are mainly generated by:

A) stationary charges

B) moving charges or electric current

C) temperature changes

D) light radiation

Solution: B

11. Electrical induction occurs when:

A) the voltage remains constant

B) the conductor length remains constant

C) the magnetic flux through an area changes

D) no charges are present

Solution: C

12. An inductor opposes rapid changes in current because it:

A) stores electric energy

B) stores magnetic energy

C) generates heat

D) neutralizes charges

Solution: B

13. An ideal transformer can:

A) convert DC voltage to higher DC voltage

B) convert AC voltage to AC voltage with different amplitude

C) permanently change the frequency of AC voltage

D) generate energy

Solution: B

14. A transient process in an electrical network refers to:

A) an instantaneous change without time dependence

B) a time-dependent transition toward steady state

C) a permanent state without variation

D) a purely mathematical concept without physical meaning

Solution: B

AC

15. A phasor represents:

A) the complete time-domain waveform

B) DC quantities only

C) only the real part of a quantity

D) a complex amplitude (magnitude and phase) at fixed frequency

Solution: D

16. The reactance of a capacitor decreases with increasing frequency:

A) increases

B) decreases

C) remains constant

D) is unpredictable

Solution: B

17. Reactive power occurs in an AC circuit when:

A) voltage and current are phase-shifted

B) voltage and current are exactly in phase

C) no energy is transferred

D) only DC current flows

Solution: A

18. A high-pass filter ideally allows:

A) low frequencies

B) no signals

C) DC only

D) high frequencies

Solution: D

19. Resonance in a resonant circuit typically means:

A) minimum impedance / maximum current (series circuit)

B) maximum damping

C) no load influence

D) instantaneous decay

Solution: A

20. A practical advantage of a symmetrical three-phase system is:

A) lower frequency

B) only two conductors are required

C) generation of a rotating magnetic field (e.g. for motors)

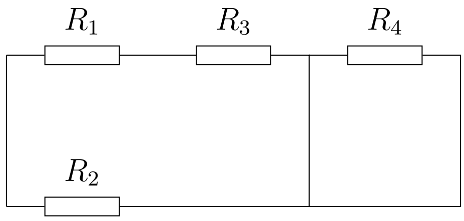
D) no balancing current possible

Solution: C

2 DC Networks

2.1 Equivalent Resistance

Given the resistor network with $R_1 = 10\,\Omega$, $R_2 = 20\,\Omega$, $R_3 = 30\,\Omega$, $R_4 = 15\,\Omega$:



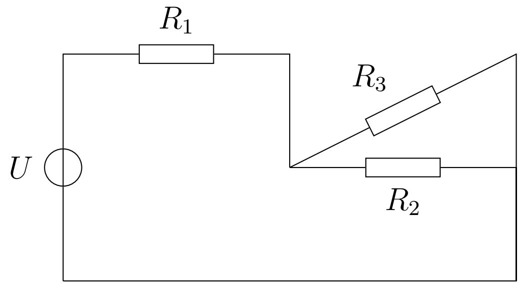
Calculate the equivalent resistance of the network.

Solution:

$$R_{12} = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \cdot 20}{30} = 6.67\,\Omega$$
$$R_{123} = 6.67 + 30 = 36.67\,\Omega$$
$$R = \frac{36.67 \cdot 15}{36.67 + 15} \approx 10.7\,\Omega$$

2.2 Network Analysis

Given the circuit with $R_1 = 6\,\Omega$, $R_2 = 12\,\Omega$, $R_3 = 4\,\Omega$ and a voltage source of 24 V:



Determine:

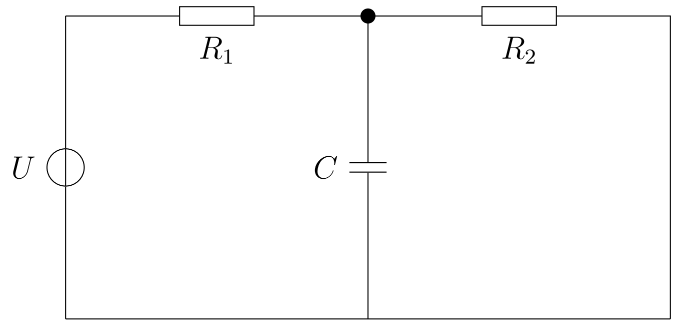
1. the total source current,
2. the branch currents through R_2 and R_3,
3. the power dissipated in R_3.

Solution:

$$R_{23} = \frac{12 \cdot 4}{16} = 3\,\Omega$$
$$R = 6 + 3 = 9\,\Omega$$
$$I = \frac{24}{9} = 2.67\,\mathrm{A}$$
$$U_{23} = I \cdot 3 = 8\,\mathrm{V}$$
$$I_2 = \frac{8}{12} = 0.67\,\mathrm{A}, \quad I_3 = \frac{8}{4} = 2\,\mathrm{A}$$
$$P_3 = I_3^2 R_3 = 16\,\mathrm{W}$$

2.3 Nodal Analysis

Given the DC network with $U = 20\,\text{V}$, $R_1 = 5\,\Omega$, $R_2 = 10\,\Omega$, $C = 100\,\mu\text{F}$:



The lower conductor is chosen as reference potential (ground). The steady-state condition ($t \to \infty$) is assumed.

Determine:

1. the node voltage at the marked node,
2. the voltage across the capacitor,
3. the current through the capacitor,
4. the energy stored in the capacitor.

Solution:

$$\frac{V_1 - 20}{R_1} + \frac{V_1 - 0}{R_2} = 0$$

$$2(V_1 - 20) + V_1 = 0$$

$$V_1 = \frac{40}{3} \approx 13.33\,\text{V}$$

$$U_C = V_1 - 0 = 13.33\,\text{V}$$

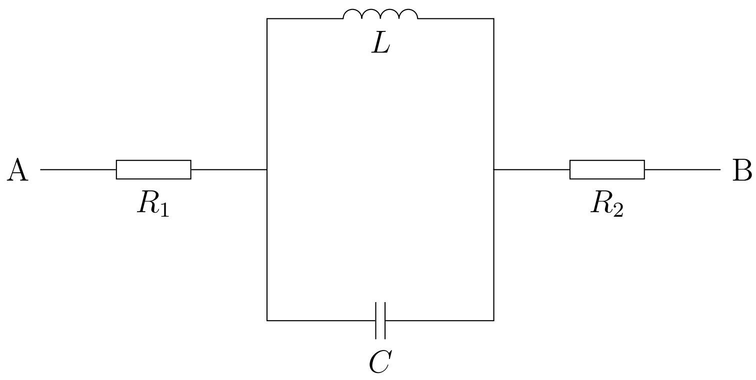
$$i_C = C\frac{du_C}{dt} = 0$$

$$E = \frac{1}{2}CU_C^2 \approx 8.9\,\text{mJ}$$

3 AC Networks

3.1 Equivalent Impedance

Given the AC network with $R_1 = 10\,\Omega$, $R_2 = 20\,\Omega$, $L = 50\,\text{mH}$, $C = 100\,\mu\text{F}$, $\omega = 1000\,\text{rad/s}$:



Determine:

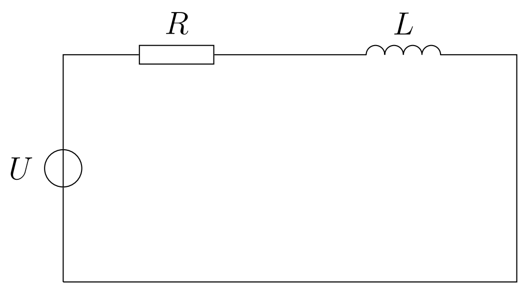
1. the complex impedances of the individual components,
2. the complex equivalent impedance between terminals A and B,
3. the magnitude and phase of the equivalent impedance.

Solution:

$$Z_R = R$$
$$Z_L = j\omega L = j \cdot 1000 \cdot 0.05 = j50\,\Omega$$
$$Z_C = \frac{1}{j\omega C} = \frac{1}{j \cdot 1000 \cdot 100 \cdot 10^{-6}} = -j10\,\Omega$$
$$Z_{LC} = \left(\frac{1}{Z_L} + \frac{1}{Z_C}\right)^{-1} = \frac{1}{j0.08} = -j12.5\,\Omega$$
$$Z = R_1 + Z_{LC} + R_2 = (30 - j12.5)\,\Omega$$
$$|Z_{\text{ges}}| = \sqrt{30^2 + 12.5^2} \approx 32.5\,\Omega$$
$$\varphi = \arctan\left(\frac{-12.5}{30}\right) \approx -22.6^\circ$$

3.2 Complex Network Analysis

Given the following network with a sinusoidal voltage source $U = 230\,\text{V}$, $f = 50\,\text{Hz}$ and the components $R = 20\,\Omega$, $L = 100\,\text{mH}$:



Determine:

1. the complex impedance Z,
2. the RMS current I,
3. the phase shift between current and voltage,
4. the power factor $\cos\varphi$,
5. the active power P,
6. the reactive power Q.

Solution:

$$\omega = 2\pi f = 314\,\mathrm{rad/s}$$

$$X_L = \omega L = 31.4\,\Omega$$

$$Z = 20 + j31.4 \Rightarrow |Z| = 37.3\,\Omega$$

$$I = \frac{230}{37.3} = 6.17\,\mathrm{A}$$

$$\varphi = \arctan\left(\frac{31.4}{20}\right) = 57.5^\circ$$

$$\cos\varphi = 0.54$$

$$P = UI\cos\varphi = 766\,\mathrm{W}$$

$$Q = UI\sin\varphi = 1180\,\mathrm{var}$$